

Supplementary Material

Drag Force

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External Flow Types

1. **Two-Dimensional:** Infinitely long with fixed cross-section (into the page)
2. **Axisymmetric:** Shaped by an axis of symmetry
3. **Three-Dimensional:** Can be symmetric to a plane rather than an axis

Flow Characteristics: Viscous vs Inviscid

Inviscid Flow

- Shear (friction) is negligible
- Typically occurs further away from walls
- Can use the Standard Bernoulli Equation (if requirements are met)

Viscous Flow

- Significant shear occurs near walls or in the wake
- Requires more complex analysis (e.g., Energy Transport Equation)

Boundary Layer

A thin region of fluid near a solid surface where:

- Viscosity effects are significant
- Friction causes flow near the surface to slow down
- Very close to the surface, fluid is almost stationary

Reynolds Number

For flow over a flat plate:

$$\text{Re}_L = \frac{\rho U L}{\mu}$$

Flow Regimes - For Flat Plates

Note: The Reynolds Number for transition can change depending on the geometry (e.g. in Ch.6 duct flow goes turbulent at a lower Re).

- **Laminar** ($\text{Re} < 10^6$): Orderly, clean, and reversible flow
- **Transitional:** Complex intermediate region
- **Turbulent** ($\text{Re} > 10^6$): Chaotic flow

Drag Force Calculation

Drag Force Formula:

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

Key Parameters

- C_D : Drag Coefficient (dimensionless)
- ρ : Fluid Density (kg/m^3)
- V : Free-stream (incoming) velocity (m/s)
- A : Reference Area (m^2)

Determine Drag Coefficient (C_D)

For Flat Plates:

- **Laminar Flow** ($Re < 10^6$):

$$C_D = \frac{1.328}{\sqrt{Re}}$$

Boundary layer thickness:

$$\frac{\delta}{x} = \frac{5}{\sqrt{Re}}$$

- **Transitional Flow:**

$$C_D \approx \begin{cases} \frac{0.031}{Re_L^{1/7}} - \frac{1440}{Re_L}, & \text{for } Re_{\text{trans}} = 5 \times 10^5 \\ \frac{0.031}{Re_L^{1/7}} - \frac{8700}{Re_L}, & \text{for } Re_{\text{trans}} = 3 \times 10^6 \end{cases}$$

- **Turbulent Flow** ($Re > 10^6$):

$$C_D = \frac{0.031}{Re^{1/7}}$$

Boundary layer thickness:

$$\frac{\delta}{x} = \frac{0.16}{Re^{1/7}}$$

Note: Real flow does not go directly from laminar to turbulent like a "light switch", it goes through a transitional period, the equation for C_D to use in this case depends on the Re_{trans} .

Area Selection

Three main types of reference areas:

1. **Frontal Area**: For thick, bluff bodies (spheres, cylinders)
2. **Planform Area**: For wide, flat bodies (flat plates, airfoils)
3. **Wetted Area**: For surface ships and barges

Steps to Solve Drag Problems (Good to have on your crib sheet)

1. Calculate Reynolds Number (Re)
2. Determine Drag Coefficient (C_D)
3. Select Appropriate Reference Area
4. Calculate Drag Force using $F_D = \frac{1}{2} C_D \rho V^2 A$
!!! Do Not Forget To Double For Plates !!! (Unless only asked for drag of one side)

Power and Drag Force

To determine the power required to overcome drag:

$$P = F_D \cdot V$$

Where:

- P : Power (in W , or converted to hp in imperial systems: $1 \text{ hp} = 745.7 \text{ W}$)
- F_D : Drag force (N or lbf)
- V : Velocity (m/s or ft/s)

For imperial units, convert power to horsepower using:

$$P_{\text{hp}} = \frac{P_{\text{ft}\cdot\text{lb/s}}}{550}$$

Pro Tips

- Reynolds number helps predict flow behavior
- Drag coefficient varies with flow regime
- Choose the right reference area for accurate calculations
- For power estimation, ensure consistent units across all variables

Additional Notes

- The drag coefficient is derived using the Buckingham Pi Theorem
- Transition between flow regimes is not a "light switch" but a gradual process
- The drag force formula is actually based on dynamic pressure (recall the $\frac{1}{2}\rho V^2$ term in Bernoulli equation)

Question 1

Problem: Consider laminar flow past the square-plate arrangements in the figure below. Compared to the drag of a single plate (F_1), how much larger is the drag of four plates together as in configurations (a) and (b)? Explain your results.

Solution: The relationship between the drag coefficient (C_D) and the Reynolds number (Re) for laminar flow is given as:

$$C_D = \frac{1.328}{\sqrt{Re}}$$

This implies that $C_D \propto L^{-1/2}$, where L is the characteristic length of the plate.

Configuration (a): Square Configuration

- The new square now has twice the length of the original plate ($2L$), and the total surface area becomes $4A_1$.
- The drag force for the square is proportional to:

$$F \propto \frac{1}{\sqrt{L}}$$

- For the four plates new square, the total drag force becomes:

$$F_a \propto \frac{4}{\sqrt{L_1/2}} = \frac{4}{\sqrt{2} \cdot \sqrt{L_1}} = \sqrt{8} \cdot F_1 \approx 2.83 \cdot F_1$$

Configuration (b): Series Plates

- The new length is now $4L$, but the total surface area remains as $4A_1$.
- Using the same relationship $F \propto \frac{1}{\sqrt{L}}$, the total drag force for four plates in series becomes:

$$F_b \propto \frac{4}{\sqrt{4L_1}} = \frac{4}{\sqrt{4} \cdot \sqrt{L_1}} = 2.0 \cdot F_1$$

Explanation of Results

- **Configuration (a):** The drag is $F_a = 2.83 \cdot F_1$. The drag is higher because each plate experiences relatively higher wall shear stress compared to configuration (b).
- **Configuration (b):** The drag is $F_b = 2.0 \cdot F_1$. Plates near the trailing edge experience less drag due to thicker boundary layers and reduced wall shear stress.

Conclusion: Drag does not quadruple because the boundary layers interact, and the effective wall shear stress decreases for plates further downstream.

Final Results:

- Configuration (a): $F_a = 2.83 \cdot F_1$
- Configuration (b): $F_b = 2.0 \cdot F_1$

Question 2

Problem Overview

A **hydrofoil** with dimensions of 1.2 ft (length) and 6 ft (width) is exposed to a **seawater flow** characterized by:

- Flow velocity: $U = 40$ ft/s,
- Density: $\rho = 1.99$ slugs/ft³,
- Kinematic viscosity: $\nu = 0.000011$ ft²/s.

Objectives:

1. Estimate the boundary layer thickness at the trailing edge.
2. Estimate the drag force under three scenarios:
 - (b) Turbulent smooth-wall flow,
 - (c) Laminar-turbulent transition flow with $Re_{\text{entry}} = 5 \times 10^5$,
 - (d) Turbulent rough-wall flow with surface roughness $\epsilon = 0.0004$ ft.

Solution

1. Reynolds Number Calculation

The Reynolds number is calculated as:

$$Re_\ell = \frac{UL}{\nu} = \frac{(40)(1.2)}{0.000011} = 4.36 \times 10^6$$

Since $Re_\ell > 5 \times 10^5$, the flow is fully **turbulent** at the trailing edge.

2. Boundary Layer Thickness ($\delta(L)$)

For turbulent flow, the maximum boundary layer thickness is given by:

$$\frac{\delta(L)}{L} = \frac{0.16}{Re_\ell^{1/7}}, \quad \delta(L) = \frac{\delta(L)}{L} \times L$$

Substituting the values:

$$\frac{\delta(L)}{L} = \frac{0.16}{(4.36 \times 10^6)^{1/7}} = 0.018, \quad \delta(L) = 0.018 \times 1.2 = 0.0216 \text{ ft}$$

Result: The boundary layer thickness at the trailing edge is:

$$\delta(L) = 0.0216 \text{ ft}$$

This is approximately **7.5 times thicker** than for a fully laminar flow.

3. Drag Force Calculations

(b) Turbulent Smooth-Wall Flow The drag coefficient for turbulent flow is:

$$C_D = \frac{0.031}{Re_\ell^{1/7}}$$

Substituting:

$$C_D = \frac{0.031}{(4.36 \times 10^6)^{1/7}} = 0.00349$$

The drag force on both sides of the hydrofoil is:

$$D = 2C_D(\rho U^2 b L)$$

Substitute the values:

$$D = 2(0.00349)(1.99)(40^2)(6)(1.2) = 80 \text{ lbf}$$

(c) Laminar-Turbulent Transition For transitional flow with $\text{Re}_{\text{entry}} = 5 \times 10^5$, the drag coefficient is modified:

$$C_D = \frac{0.00349}{(4.36 \times 10^6)^{1/7}} = 0.00316$$

The drag force becomes:

$$D = 2C_D(\rho U^2 b L) = 2(0.00316)(1.99)(40^2)(6)(1.2) = 72 \text{ lbf}$$

(d) Turbulent Rough-Wall Flow The roughness ratio is:

$$\frac{\epsilon}{L} = \frac{0.0004}{1.2} = 0.000333$$

From Fig. 7.6, this roughness places the flow in the **fully rough regime**. The drag coefficient is:

$$C_D = (1.89 + 1.62 \log_{10}(3000))^{-2.5}, \quad \text{where } 3000 = \frac{\epsilon}{\nu/U}$$

Substituting:

$$C_D = (1.89 + 1.62 \log_{10}(3000))^{-2.5} = 0.00644$$

The drag force is:

$$D = 2C_D(\rho U^2 b L) = 2(0.00644)(1.99)(40^2)(6)(1.2) = 148 \text{ lbf}$$

Summary of Results

- Boundary layer thickness at the trailing edge: $\delta(L) = 0.0216 \text{ ft}$.
- Drag forces:
 - Smooth-wall turbulent flow: $D = 80 \text{ lbf}$,
 - Laminar-turbulent transition: $D = 72 \text{ lbf}$,
 - Rough-wall turbulent flow: $D = 148 \text{ lbf}$.
- **Effect of Roughness:** Small roughness ($\epsilon = 0.0004 \text{ ft}$) nearly **doubles the drag**, and trailing-edge flow separation could further increase drag by a factor of 2.

Question 3

Problem Overview

We are tasked with estimating the horsepower required to drive a pickup truck at a speed of 55 mph (80.7 ft/s) under two scenarios:

1. For the clean truck (no sign).
2. With a 3-by-6-ft sign installed, as shown in Fig. P7.88.

Additionally, the rolling resistance is given as 150 lbf at sea level.

Given Data

For the clean case, the following values are provided:

- Clean drag-area (CDA): 35 ft^2 ,
- Air density: $\rho = 0.00238 \text{ slug/ft}^3$,
- Dynamic viscosity: $\mu = 3.72 \times 10^{-7} \text{ slug/ft-s}$,
- Rolling resistance: $F_{\text{rolling}} = 150 \text{ lbf}$,
- Velocity: $V = 80.7 \text{ ft/s}$ (55 mph).

For the case with the sign added:

- Effective height-to-base ratio: $h/b = 2.0$,
- Drag coefficient for the sign: $C_D \approx 1.19$ (from Table 7.3).

Solution

(a) Clean Truck Horsepower

The total drag force on the truck is the sum of the rolling resistance and aerodynamic drag. The aerodynamic drag is calculated using the formula:

$$F = F_{\text{rolling}} + C_D A \left(\frac{\rho}{2} \right) V^2$$

Substitute the given values for the clean truck:

$$F = 150 + 35 \left(\frac{0.00238}{2} \right) (80.7)^2$$

First, calculate the aerodynamic drag:

$$F_{\text{aero}} = 35 \times \left(\frac{0.00238}{2} \right) \times (80.7)^2 = 421 \text{ lbf}$$

So, the total drag force is:

$$F_{\text{total}} = 150 + 421 = 421 \text{ lbf}$$

The horsepower required to overcome this drag force is given by:

$$\text{Horsepower} = \frac{F \times V}{550}$$

Substitute the values:

$$\text{Horsepower} = \frac{421 \times 80.7}{550} = 62 \text{ hp}$$

Result for (a): The horsepower required for the clean truck is 62 hp.

(b) Horsepower with the Sign Installed

For the case with the 3-by-6-ft sign installed, the drag force increases due to the additional drag caused by the sign. The drag force is calculated as:

$$F = F_{\text{rolling}} + C_D A \left(\frac{\rho}{2} \right) V^2 + C_D \left(\frac{\rho}{2} \right) V^2 A_{\text{sign}}$$

Substitute the values for the truck with the sign:

$$F = 421 + 1.19 \left(\frac{0.00238}{2} \right) (80.7)^2 (6)(3)$$

First, calculate the additional drag force from the sign:

$$F_{\text{sign}} = 1.19 \times \left(\frac{0.00238}{2} \right) \times (80.7)^2 \times (6)(3) = 166 \text{ lbf}$$

So, the total drag force is:

$$F_{\text{total}} = 421 + 166 = 587 \text{ lbf}$$

The horsepower required to overcome this total drag force is:

$$\text{Horsepower} = \frac{F_{\text{total}} \times V}{550}$$

Substitute the values:

$$\text{Horsepower} = \frac{587 \times 80.7}{550} = 86 \text{ hp}$$

Result for (b): The horsepower required with the sign installed is 86 hp.

Summary of Results

- **(a) Clean truck:** The required horsepower is 62 hp.
- **(b) With the sign installed:** The required horsepower is 86 hp.

Question 4

[Note: This is easier than Q3 but shows what happens if we look at partially immersed object (e.g. boats) instead]

Problem: A ship is 125 m long and has a wetted area of 3500 m². Its propellers can deliver a maximum power of 1.1 MW to seawater at 20°C. If all drag is due to friction, estimate the maximum ship speed in knots.

Solution: For seawater at 20°C, take:

$$\rho = 1025 \text{ kg/m}^3, \quad \mu = 0.00107 \text{ kg/m-s.}$$

Step 1: Calculate Reynolds Number

The Reynolds number based on the ship length ($L = 125 \text{ m}$) is given by:

$$Re_L = \frac{\rho UL}{\mu} = \frac{1025 \cdot V \cdot 125}{0.00107} \approx 1.2 \times 10^8 \cdot V.$$

This value confirms that the flow is turbulent.

Step 2: Determine Drag Coefficient (C_D)

For turbulent flow, the drag coefficient is approximated by:

$$C_D = \frac{0.031}{Re_L^{1/7}}.$$

Substitute $Re_L = 1.2 \times 10^8 \cdot V$:

$$C_D = \frac{0.031}{(1.2 \times 10^8 \cdot V)^{1/7}} \approx \frac{0.031}{V^{1/7}} = 0.00217 \cdot V^{-1/7}.$$

Step 3: Power Balance

The ship's drag force (F) and power (P) are related by:

$$P = F \cdot V = \frac{1}{2} C_D \rho V^2 A \cdot V.$$

Substituting $C_D = 0.00217 \cdot V^{-1/7}$, wetted area $A = 3500 \text{ m}^2$, and maximum power $P = 1.1 \times 10^6 \text{ W}$:

$$1.1 \times 10^6 = [0.00217 \cdot (1025) \cdot V^2 \cdot 3500] \cdot V^{-1/7}.$$

Simplify:

$$1.1 \times 10^6 = 7.8 \times 10^6 \cdot V^{20/7}.$$

Step 4: Solve for Ship Speed (V)

Rearrange to isolate $V^{20/7}$:

$$V^{20/7} = \frac{1.1 \times 10^6}{7.8 \times 10^6} \approx 0.141.$$

Take the 7/20 power to solve for V :

$$V \approx 7.2 \text{ m/s.}$$

Step 5: Convert to Knots

Convert V from m/s to knots ($1 \text{ m/s} = 1.9438 \text{ knots}$):

$$V \approx 7.2 \cdot 1.9438 \approx 14 \text{ knots.}$$

Final Answer: The maximum ship speed is approximately 14 knots.